Sample Defense Questions

1. What is a brain stroke?
   - A brain stroke occurs when blood flow to the brain is interrupted, causing damage due to lack of oxygen.
2. Why is early stroke detection important?
   - It helps reduce brain damage and improves patient recovery and survival.
3. Which imaging technique did you use?
   - CT scan images were used in this study.
4. Why did you choose CT scans?
   - Because they are fast, widely available, and suitable for emergency diagnosis.
5. What is the main objective of your research?
   - To develop an automated system for detecting brain stroke using deep learning.
6. Which models did you use in your study?
   - VGG16, ResNet, DenseNet-201, and EfficientNet.
7. What is deep learning?
   - A type of machine learning that automatically learns patterns from data.
8. What is a CNN?
   - A Convolutional Neural Network used for image analysis tasks.
9. Why are CNNs used in medical imaging?
   - Because they can automatically extract important features from images.
10. What is transfer learning?
    - Using a pre-trained model and adapting it to a new task.
11. What preprocessing steps did you apply?
    - Resizing, normalization, noise reduction, and data augmentation.
12. Why is data augmentation important?
    - It increases dataset size and improves model generalization.
13. What is class imbalance?
    - When one class has more samples than another.
14. How did you handle class imbalance?
    - By applying data augmentation and balancing techniques.

15. What evaluation metrics did you use?
    - Accuracy, precision, recall, and F1-score.
16. Why is recall important in your study?
    - Because missing a stroke case can be very dangerous.
17. What is a confusion matrix?
    - A table that compares predicted and actual results.
18. Which model performed the best?
    - VGG16 performed the best overall.
19. What are the limitations of your study?
    - Limited dataset, binary classification, and lack of interpretability.
20. What is the future scope of your research?
    - Using larger datasets, multi-modal data, and explainable AI.

21. What type of problem is your model solving?
    - It is a binary classification problem (stroke vs normal).
22. What is the input to your model?
    - CT scan images of the brain.
23. What is the output of your model?
    - A prediction indicating whether the image is stroke or non-stroke.
24. Why did you compare multiple models?
    - To identify which architecture performs best under the same conditions.
25. What is overfitting?
    - Overfitting occurs when a model performs well on training data but poorly on unseen data.
26. How did you prevent overfitting?
    - By using data augmentation, validation, and early stopping.
27. What optimizer did you use?
    - Adam optimizer.
28. What loss function did you use?
    - Binary cross-entropy loss.

29. What is accuracy?
    - The percentage of correct predictions out of total predictions.
30. What is precision?
    - The proportion of correctly predicted positive cases out of all predicted positives.
31. What is recall?
    - The proportion of correctly predicted positive cases out of actual positives.
32. What is F1-score?
    - The harmonic mean of precision and recall.
33. What is a false negative?
    - A stroke case incorrectly predicted as normal.
34. Why are false negatives dangerous?
    - Because missing a stroke can delay treatment and harm the patient.
35. What is normalization?
    - Scaling pixel values to a standard range to improve model performance.
36. What is resizing in preprocessing?
    - Adjusting all images to a fixed size for model input.
37. Why is GPU used?
    - To speed up model training.
38. What is inference time?
    - The time taken by the model to make a prediction.
39. What is generalization?
    - The ability of a model to perform well on new, unseen data.
40. Why is your system useful in healthcare?
    - It helps doctors by providing fast and reliable decision support.
41. Can your model replace doctors?
    - No, it is designed to assist, not replace medical professionals.
42. What is the main challenge in your dataset?
    - Limited size and class imbalance.
43. What is feature extraction?
    - The process of identifying important patterns from images.

44. What is fine-tuning?
    - Adjusting pretrained model layers for a specific task.
45. Why is EfficientNet efficient?
    - Because it balances depth, width, and resolution.
46. What is a residual connection in ResNet?
    - A shortcut connection that helps in training deep networks.
47. What is DenseNet’s advantage?
    - Better feature reuse through dense connections.
48. Why is VGG16 simple?
    - Because it uses a straightforward sequential architecture.
49. What is ROC curve?
    - A graph showing model performance across different thresholds.
50. What is the main contribution of your research?
    - A fair comparison of multiple CNN models for stroke detection.